

Named Entity Recognition, Sentiment Analysis, and Topic Classification: A Comparative Study

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1. Introduction

Proper nouns such as person names, locations, and organizations carry important semantic information in text. Named-Entity Recognition and Classification (NERC) addresses this by automatically detecting and classifying such entities. We compare the performance of spaCy and BERT on unseen test data in this project. Sentiment analysis identifies the emotional polarity of a text. The process of identification is crucial in NLP and has heavy applications in social media. Thus, in this project, the model performs sentence level classification to categorize unseen sentences into three sentiments: negative, positive or neutral. Topic classification is the task of automatically assigning a predefined categorical label representing the main subject to a given text. In this project, we perform sentence-level classification to categorize unseen sentences into three domains: Movie, Book, or Restaurant.

2. Data

2.1 Train Data

2.1.1 Topic Classification

We trained our topic classifier with three datasets from Kaggle, each corresponding to one of the three labels in the testing data:

- IMDB Dataset of 50K Movie Reviews:** This dataset is owned by Lakshminpathi N. and holds approximately 50,000 IMDB movie reviews alongside corresponding sentiment labels.
- Yelp Restaurant Reviews:** This dataset is owned by Faruk Alam and contains approximately 20,000 reviews alongside meta-data collected across more than 15 years.
- Amazon Book Reviews:** This dataset is owned by Mohamed Bekheet and holds approximately three million book reviews for approximately 212,404 unique book.

These datasets were chosen based on their proximity in tone, numbers of samples (>19,000 each) and domains matching the task's labels.

2.1.2 Sentiment Analysis

We trained our model on course material; namely the airline tweets supplied in lab three. This dataset contained three directories:

- Airline Tweets:** 1490 positive, 1750 negative, and 1515 neutral .txt tweets.

See the data under data/raw/train in the linked repository.

2.2 Test Data

2.2.1 Sentiment & Topic (Sentiment-topic-test.tsv)

- Source: VU Text Mining Canvas
- Size: 10 sentences
- Label Distribution: 4 positive, 3 negative, 3 neutral; 4 Movie, 3 Book, 3 Restaurant

2.2.2 NERC (NER-test.tsv)

- Source: VU Text Mining Canvas
- Size: 10 sentences, 17 named entities
- Label Distribution: 6 PER, 4 ORG, 4 LOC, 3 MISC

3. Methodology

3.1 NERC — System A (spaCy)

We employed the pretrained spaCy model en_core_web_sm for named entity recognition. The model identifies entities using statistical features learned from annotated training data, including token shape, capitalization patterns, prefixes and suffixes, surrounding words and part-of-speech information. It predicts entity spans and types such as PERSON, ORG and LOC. Unlike System B (BERT), spaCy relies on a lightweight statistical pipeline optimized for speed and efficiency rather than deep contextual transformer representations. The model was applied directly to the test sentences and the predicted entities were converted into BIO tags for evaluation.

3.2 NERC — System B (e.g. BERT / transformer)

We employed the pretrained model dslim/bert-base-NER, which is based on the BERT model (Devlin et al., 2019) fine-tuned on CoNLL-2003 for NER and available through Hugging Face. BERT processes text bidirectionally, capturing both left-to-right and right-to-left context simultaneously via its transformer architecture. Input tokens are handled through subword tokenization. Unlike System A (spaCy), no manual feature engineering is required, as the model leverages full contextual representations directly.

3.3 Sentiment Analysis — cardiffnlp RoBERTa Transformer

Large pre-trained language models significantly optimize sentiment analysis within NLP (Devlin et al., 2019). Transfer learning is employed for the task by fine-tuning the cardiffnlp/twitter-roberta-base-sentiment, a transformer model pretrained on 124 million tweets (Barbieri et al., 2020). Data pre-processing is carried out prior to tokenisation, where user mentions and urls are normalized to @user, and http respectively. Token length is set to 128; a length sufficient to capture all semantics at the maximum tweet length. The model architecture is comprised of the pretrained RoBERTa encoder, followed by a linear classification head. The [CLS] vector is mapped to the three sentiment classes aforementioned. The feature representations are tensors of dimension 768, and fine tuning is done over four epochs, using the AdamW optimizer and a linear learning rate warmup -over 10% of training-

-steps: ($lr = 2 \times 10^{-5}$, weight decay = 0.01). The model selection will be based on validation accuracy and a final evaluation on the test set will evaluate the model.

3.4 Topic Classification

We used a pretrained contextual transformer model, specifically bert-base-cased, which was instantiated via the SimpleTransformers library. As discussed in lectures, traditional Bag-of-Words (BoW) models suffer from data sparseness and struggle with out-of-vocabulary words. We bypassed this by using BERT (Devlin et al., 2019). Similarly to Sun et al. (2020), we map the [CLS] vector from BERT's last layer onto classes and reduced proneness to catastrophic forgetting during fine-tuning by choosing a low learning rate of $4 \cdot 10^{-6}$. BERT is a bidirectional transformer and uses the [CLS] token as a sentence-level vector embedding. Each of the datasets was reduced to a single text column, truncated to 1000 samples, and augmented with an integer column identifying the source (equivalently, class). The three datasets were concatenated vertically, shuffled and split into train (90%) and development (10%) sets.

4. Results

4.1 NERC

System A (spaCy) achieved an overall F1 score of 0.86 with strongest performance on ORG entities. System B (BERT) achieved a higher overall F1 score of 0.91, and outperformed spaCy across most entity categories, particularly LOC and MISC entities.

Label	System A (spaCy)			System B (BERT)		
	P	R	F1	P	R	F1
PER	0.67	0.50	0.57	0.84	0.71	0.77
ORG	0.75	1.00	0.86	1.00	0.88	0.93
LOC	1.00	0.25	0.40	0.90	1.00	0.95
MISC	0.00	0.00	0.00	1.00	1.00	1.00
Overall	0.89	0.85	0.86	0.94	0.90	0.91

Table 1: NERC results for System A and System B.

4.2 Sentiment Analysis

Class	Precision	Recall	F1-score	Support
Negative	1.00	0.67	0.80	3
Neutral	0.75	1.00	0.86	3
Positive	0.75	0.75	0.75	4
Accuracy			0.80	10
Macro avg	0.83	0.81	0.80	10
Weighted avg	0.82	0.80	0.80	10

Table 2: Sentiment Analysis results (twitter-roberta)

4.3 Topic Classification

Our topic classifier correctly predicted classes for all ten test samples. Thus, for every pair of class and precision / recall / F1 score, said classifier exhibits the optimal value of 1.0.

	P	R	F1	S
Book	1.0	1.0	1.0	2.0
Movie	1.0	1.0	1.0	5.0
Restaurant	1.0	1.0	1.0	3.0
accuracy	1.0	1.0	1.0	1.0
macro avg	1.0	1.0	1.0	10.0
weighted avg	1.0	1.0	1.0	10.00

Table 3: Topic classification results.

5. Error Analysis — NERC

System A (spaCy) showed weaker performance on LOC and MISC entities, resulting in a lower overall F1 score than System B. In particular, spaCy failed to correctly identify any MISC entities and struggled with location recall. For System B (BERT), two main error types were identified. The first was entity type misclassification: “Blauwbrug” was predicted as LOC instead of ORG, likely due to the model being trained on newswire data (CoNLL-2003) rather than review-domain text. The second was boundary detection failure. In “Dame Maggie Smith” and “Mr. Kruno”, the honorifics were labeled O, causing the following tokens to be incorrectly tagged. Despite these errors, System B achieved strong overall performance with only 5 misclassifications out of 214 tokens. The results suggest that contextual information helped BERT make more accurate predictions, while System A's lighter statistical approach was less effective on challenging entity types.

Sentiment Analysis - Confusion Matrix

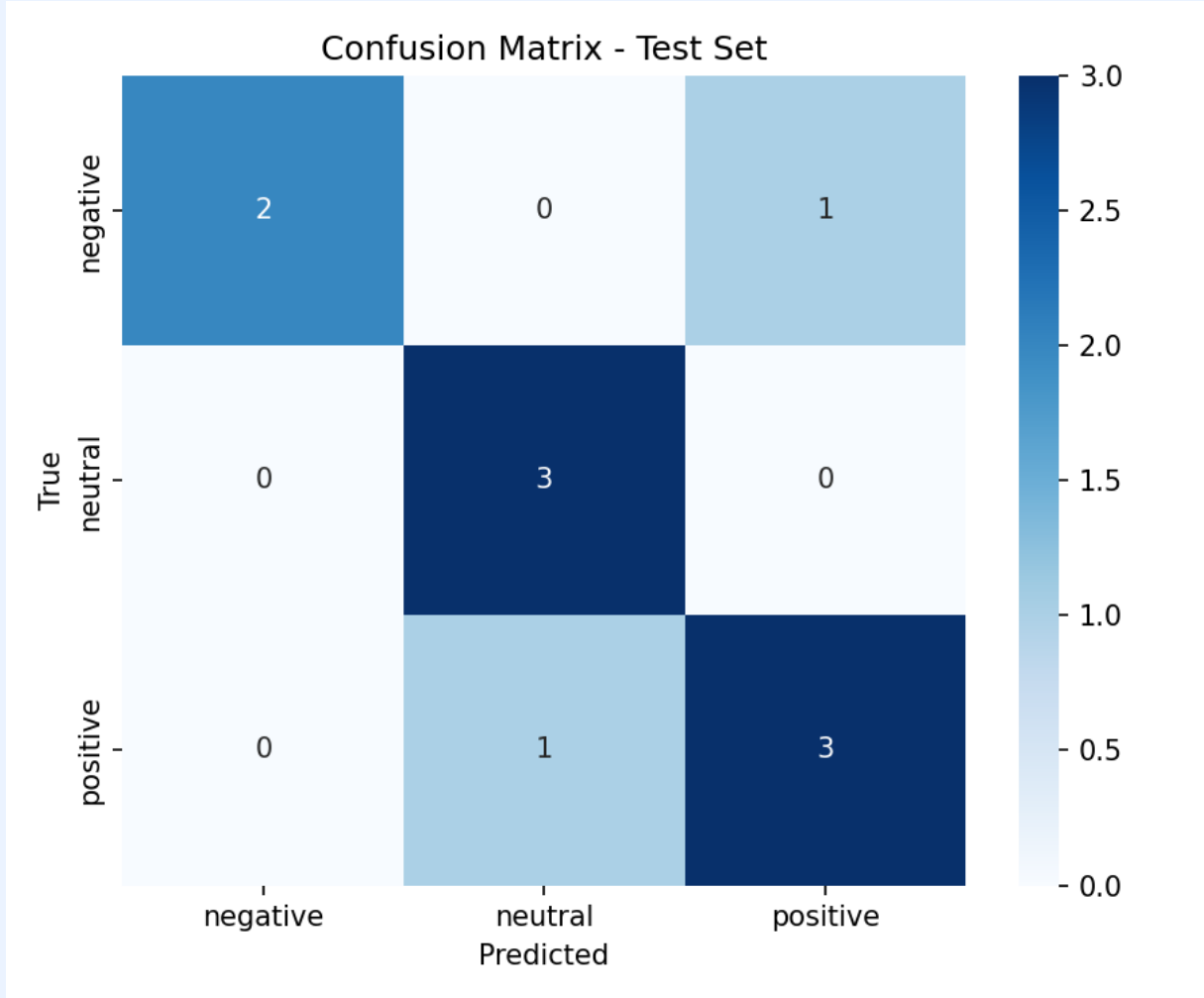


Figure 1: Confusion matrix for sentiment model evaluation

6. Discussion

As seen in (2), the fine tuned model achieves a weighted F1 score of 83% on the test set, showing it is able to generalize well despite a heavy domain mismatch on the test-train data. Precision for the negative sentiment is perfect (100%), but comparatively, the lowest recall (67%)- this is attributable to the fact that the model is highly confident, but misses a large portion of TN instances. The neutral sentiment achieves the highest overall score (F1=86%), with perfect recall. Suggesting the neutral sentiment is a fallback for ambiguous cases. The positive sentiment is the weakest class (F1=75%), with balanced precision and recall. The lower scores reflect the domain disparity. Two primary error patterns emerge from the confusion matrix (1). First, positive tweets are frequently misclassified as neutral. This reflects a conservative bias in the model — when positive sentiment is expressed subtly or indirectly, for example "In conclusion, my review of this book would be: I like Jane Austen and understand why she is famous.", the model hedges toward neutral rather than committing to a positive prediction. Second, a smaller number of negative tweets are misclassified as positive, most likely attributable to sarcasm and implicit negativity — tweets such as "I always have loved English novels, but I just couldn't get into this one." carry surface-level positive vocabulary that misleads the classifier despite the clearly negative intent.

Our topic classifier achieved an optimal macro F1 score of 1.0 on the Sentiment-topic-test.tsv dataset. However, the test set is highly constrained (only 10 sentences) and lacks the volume required to reveal edge cases. Furthermore, the domains (Movie, Book, Restaurant) have relatively distinct, little-overlapping vocabularies. Because the classes are semantically distant, the classification task is relatively trivial for a transformer model like BERT. The model likely made its decisions based on the vocabulary used instead of complex semantic understanding.

7. Conclusions & Limitations

For NERC, System A (spaCy) achieved an overall F1 score of 0.86, with its strongest performance on ORG entities (F1 = 0.86). However, performance was weaker on LOC entities (F1 = 0.40) and the model failed to correctly identify MISC entities in the test set. These results suggest that while spaCy provides a fast and efficient NER solution, it is less effective than transformer-based approaches when handling more challenging entity categories. System B (BERT) outperformed System A (spaCy), demonstrating that contextual representations are highly effective for named entity recognition. However, two limitations were observed. The model showed domain mismatch when handling non-English proper nouns, and struggled with honorific boundary detection. These findings reflect known constraints of models trained solely on newswire data (CoNLL-2003) when applied to review-domain text. Future work could address this through fine-tuning on review-domain NER data and incorporating title and honorific gazetteers. The topic classifier implementation heavily prioritized a short training duration. While the classifier correctly labeled every training samples, other classifiers might outperform it by increasing the number of samples, the number of epochs or the maximum sequence length. In addition, the task's exact source or distribution is unknown; under knowledge thereof, prediction accuracy could be improved by selecting training data whose underlying distribution more closely matches the task's underlying distribution. Future work could further treat BERT configurations like the learning rate and batch size as hyperparameters to be optimized, as opposed to constants. Finally, populating the test set with further sentences and ensuring more even support between classes can increase the accuracy of our performance evaluation. The sentiment analysis model achieved a weighted F1 of 0.80 on the test set, demonstrating reasonable generalisation despite a significant domain mismatch between airline tweet training data and the general-domain test set. The primary limitation is the small test set of only 10 samples, which limits the statistical reliability of the reported scores- a single misclassification shifts metrics considerably. Moreover, the model struggles with implicit negativity and sarcasm. It shows a conservative bias toward neutral predictions in ambiguous cases. Given more time, performance could be improved by training on in-domain data matching the test distribution, applying class weighting to address potential imbalance, and incorporating a sarcasm detection preprocessing step.

8. References

- Devlin et al. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, 4171–4186. <https://aclanthology.org/N19-1423/>
- Sun, Chi, Xipeng Qiu, Yige Xu, and Xuanjing Huang. "How to Fine-Tune BERT for Text Classification?" arXiv [Cs.CL]. 14 May 2019. arXiv. <https://doi.org/10.48550/arXiv.1905.05583>.
- Barbieri, F., Anke, L. E., & Camacho-Collados, J. (2022, June). XLM-T: Multilingual language models in Twitter for sentiment analysis and beyond. In Proceedings of the thirteenth language resources and evaluation conference (pp. 258-266).

9. Division of Work

	Coding	Analysis	Poster
Minhye Kang	NERC System B (BERT)	NERC System B Results Evaluation & Error Analysis & Conclusions	Sections 3.2, 4.1, part of 1, 5, 7, and 8
Sebastian Nagler	Topic Classification	Result Analysis: Topic Classification	Sections 3.4, 4.3, part of 1, 2, and 7
Boglarka Ori	NERC System A (spaCy)	NERC Results Evaluation & Error Analysis & Conclusions	Sections 3.1, 4.1, part of 5 and 7
Louis Battle	Sentiment Analysis	Result Analysis: Sentiment Analysis	Sections 3.3, 4.2, part of 1, 2, 6, and 7.

10. Source Code

Sentiment Analysis: <https://github.com/l-battle/ta-sentiment-analysis>
Remaining pipeline (also including sentiment analysis) submitted as .zip file on canvas